

Master Learning Plan v7

Medical-Imaging + Geometric/Molecular-ML — Research-Engineer Readiness — Master Learning Plan (v7)

Owner: MD Hashim · **Window:** opens now (mid-Aug 2026); paced, not fixed · **Total effort:** ~1,360 focused hours (down from v6's ~1,950) · **Structure:** two blocks + three continuous threads (DSA/C++, implement-by-hand ML, research output) + an application overlay that starts **now**.

This is v6 **re-engineered, not rewritten**. Every resource and URL from v6 is preserved (see the Resource index). What changed is *sequencing, hours, weighting, and positioning* — to reach the **earliest realistic job-readiness** for the actual near-term target: **Research Engineer / applied ML at a life-sciences AI lab** (RS is PhD-gated and long-term). The plan now: compresses the reactivation-heavy early phases (you already ship medical segmentation and quantum chemistry — those are refreshes, not re-learning); compresses quantum and moves it *off* the critical path; cuts the legacy "generic senior CV engineer" block and pulls the life-sciences **video** piece *up* next to imaging; right-sizes DSA/system-design to the research-lab loop; de-duplicates the threads against the cross-cutting habits; rennumbers the capstones; and adds three sections v6 lacked — a **PhD-later decision guide**, a **publication/portfolio strategy**, and a **risk register**.

Changelog (v6 → v7) — read this first

Area	v6	v7
Title / positioning	"Senior AI/CV + Life-Sciences-Research"	" Medical-Imaging + Geometric/Molecular-ML — Research-Engineer Readiness. " CV-beyond-medical is an honest <i>growth area</i> , not seniority. Near-term target = RE/applied ML , not RS
Total hours	~1,950	~1,360 (reactivation compression + legacy cut + right-sized DSA + de-dup)
Critical path to first job	Tier 0 after ~420 hrs (~early Nov)	Tier 0 eligible now ; capstone-strengthened in ~weeks. Apply in parallel
Quantum (Phase 4)	~275 hrs, mid-Block-1, on the path	~100 hrs, off the critical path ; folded into the molecular-ML bridge (Capstone 3 absorbs old 3b)
Phase 8 "CV Expert"	230 hrs (classical/3D/SLAM/NeRF/edge-C++ + video)	Legacy classical/3D/edge → optional literacy (~30 hrs) ; video/temporal re-sequenced up into new Phase 2B next to imaging
DSA / system design	Thread T ~4 hrs/wk + P9 55 hrs + generic DDIA/6.824	Thread T ~2.5 hrs/wk , crescendo DSA 55→35, system design → <i>ML</i> system design. RE loop = 2 medium-hard coding rounds
Threads vs habits	Thread M ≈ habit #3; Thread R ≈ habits #6/#7 (double-counted)	Merged ; hours counted once
Capstones	C1, C3b, C3, C4, C2, C5 (scrambled)	C1 seg → C2 video → C3 molecular (flagship, absorbs quantum) → C4 agent (optional) ; C5 dropped
Weekly load	Fixed cap (34-38 / 28-32 hrs/wk)	Per-phase hours + flexible pace ; 45 hrs/wk treated as a <i>sprint</i> , then throttle
Framework	PyTorch-only	+ light JAX (relocation labs run JAX)
New sections	—	PhD-later decision guide · Publication & portfolio strategy · Risk register
Everything else from v6	—	Preserved (all URLs, COI guardrails, resource index, capstone datasets, company ladder)

The program at a glance

v7 keeps v6's two-block spine and light continuous scaffold, but re-sequences the critical path so you interview early and often.

Block 1 — Depth, foundations & the spike (opens now, ~830 hrs). A rapid articulation brush-up, then a *compressed* reactivation of math foundations and your medical-CV domain (you already ship this), then the **imaging+video** consolidation (fallback leg, near-zero new hours), then DL theory → a *compressed, off-path* quantum refresh → production → the **geometric/molecular-ML spike** (the differentiator for the relocation target). Paced flexibly; sprint the early path.

Block 2 — Breadth + research/lab-interview readiness (~245 hrs + interviewing). GenAI & agents (domain-focused, *lighter*), an optional CV-breadth literacy appendix, then a research-and-lab interview crescendo (DSA at RE weight, implement-by-hand ML, ML system design, research job-talk, behavioral).

Threads (parallel, from the Phase-1 self-test gate). DSA + C++ (light), implement-by-hand ML (the highest-ROI thread — the RE coding round *is* implement-from-scratch), and research output & networking (carries your live applications and seeds a later PhD).

Application overlay (from now). You are already eligible for the India-fallback tier. Enter the funnel now at Tier 0; each capstone unlocks the next tier while the plan continues underneath.

Why this order, and why it's faster than v6

You are not a beginner in your two strengths, so the plan stops treating you like one. v6 budgeted ~420 hours of foundations + a re-teaching of medical segmentation *before* your first application — but segmentation is your production domain and most of the early math is a *refresh*, not new acquisition. v7 compresses that reactivation and starts your applications in parallel, because for the India-fallback tier your **eligibility exists today**; the capstone strengthens the case, it doesn't gate it. The stated pain point — articulating fundamentals to a panel — is a *retrieval* problem, cured by targeted reactivation plus spoken drills, not by re-watching 40 hours of linear-algebra lectures.

Quantum comes off the critical path. You ship VQE/ADAPT-VQE/DMET-VQE in production, and the relocation targets (Isomorphic/DeepMind-science) hire on DL/GNN/generative, not quantum computing. So quantum is compressed to the minimum that (a) lets you defend your existing work and (b) powers the molecular-ML bridge — and it runs *later/parallel*, never blocking a target-relevant tier.

The spike is protected and slightly deepened. Geometric/equivariant ML for molecules — bridged from your quantum chemistry — is the one thing that separates you from the field for the relocation target, and it seeds a PhD. It's the one place to *spend* freed hours, not save them.

Muscle-memory threads still start at Month ~2 (the Phase-1 self-test), not today. Block 1's opening is the most focus-hungry stretch; diluting it with daily disciplines is a bad trade, and these skills reward consistency over a long horizon more than front-loaded volume. Thread M (implement-from-scratch) is the cheapest, best-validated edge you have — DeepMind's RE coding round literally asks candidates to implement a loss function (e.g. Focal Loss) from a blank file, which is exactly what you'll have been drilling weekly.

Timeline, pace & the critical path

Weekly effort is flexible. You've stated ~45 hrs/week capacity on top of a full-time role. Treat that as a **sprint** setting for the near-term critical path (the ~4-12 weeks to Tier-0 and a strong start on the spike) — ~90-hour weeks are sustainable for weeks, not for a year. **Then throttle to ~25-30 hrs/week** for the long tail; burnout in month five costs more than any schedule it buys. The self-tests and capstones are the real gates; dates are planning aids.

Elapsed time under three sustainable paces (window opens now):

Milestone	Focused hrs	@20 hrs/wk (lighter)	@30 hrs/wk (heavier-sustainable)	@45 hrs/wk (sprint only)
Tier-0 eligible to apply	~0 (now)	now	now	now
Tier-0 capstone-strengthened	~180	~9 wks	~6 wks	~4 wks
Tier-2 RE loop credible	~525 core	~26 wks (~mid-Feb 2027)	~18 wks (~late-Dec 2026)	~12 wks (~early-Nov 2026)
Full program (all enrichment)	~1,360	~68 wks	~45 wks	~30 wks

Critical path to Tier 0 (India medical-imaging + accessible) ≈ ~180 hrs: P0 brush-up (35) → P1 high-leverage slice (~55) → P2 packaging + Capstone-1 (~90). *Apply in parallel now.*

Critical path to Tier 2 (Isomorphic/DeepMind RE — relocation) ≈ ~525 hrs core: P0 (35) → P1 full (120) → P3 (105) → **P6 spike + flagship capstone (175)** → P5 ML-system-design slice (~30) → P9 job-talk + DSA/implement-from-scratch crescendo (~60), with **Thread M running throughout**. Quantum, GenAI, and legacy-CV are *not* on this path.

Phase summary (the whole plan at a glance)

Block 1 — Depth, foundations & the spike · ~830 hrs

Phase	Focus	Effort (v6→v7)	Unlocks
0	Rapid brush-up — stats, ML & DL fundamentals (articulation)	45 → 35	—
1	Foundations — mathematics + neural nets from first principles	190 → 120	—
2	Applied CV & Medical Image Segmentation (+ Capstone 1)	185 → 110	Tier 0
2B	Imaging & Video (life-sciences-native) (+ Capstone 2) (<i>re-sequenced from v6 P8</i>)	— → 95	Tier 1 (imaging)
3	Deep Learning theory & modern architectures	150 → 105	Tier 1 (broad)
4	Quantum & QML — <i>compressed, off the critical path</i>	275 → 100	—
5	Full-stack / production AI systems	160 → 105	—
6	Geometric/molecular ML — the spike (+ Capstone 3 , flagship)	150 → 175	Tier 2

Block 2 — Breadth + research/lab-interview readiness · ~245 hrs + interviewing

Phase	Focus	Effort (v6→v7)	Unlocks
7	GenAI & Agentic AI (domain-focused, <i>lighter</i>) (+ Capstone 4 , optional)	150 → 95	Tier 2 broadened
8	CV-breadth literacy (classical/3D/edge) — <i>optional appendix</i>	230 → ~30	—
9	Research + Big-Tech Interview Crescendo — DSA (RE weight), implement-by-hand ML, ML system design, job-talk, behavioral	150 → 115	Tier 3
—	Active interviewing + maintenance + buffer	—	—

Threads T + M + R · ~5 hrs/wk combined (down from ~7.5) · start at the Phase-1 self-test gate (~Month 2)

Grand total ≈ **~1,360 focused hours**. Specialized math (optimization, information theory, group theory beyond the Phase-1 core) is threaded into the phase that uses it; hours are included above.

Threads (the muscle-memory + output scaffold)

All three start at the **Phase-1 self-test** (~Month 2) and run continuously. Do them first thing, before the day's phase work, while your head is fresh. On heavy weeks, dial the combined thread load to ~2-3 hrs — consistency beats volume.

Thread T — DSA + C++ · ~2.5 hrs/week (re-weighted down from ~4)

RE loops need medium-hard DSA fluency (typically **two** coding rounds), not a FAANG-hard grind. C++ is nice-to-have (DeepMind CoderPad quirk; edge work) but not core for a JAX/PyTorch science lab — so the C++ stage is shorter.

Stage T1 — C++ essentials (compressed). Enough modern C++ to solve problems in it and reason about hardware: types, references/pointers, RAII, the STL, classes, templates, smart pointers.

- learncpp.com (sections 1-17) — <https://www.learncpp.com/>
- The Cherno — C++ series — <https://www.youtube.com/@TheCherno>
- Reference — <https://en.cppreference.com/>

Stage T2 — DSA, sustained. ~2-3 problems/week, spaced, pattern-based.

- NeetCode — pattern roadmap; work the NeetCode 150 — <https://neetcode.io/>
- LeetCode — the problem bank — <https://leetcode.com/>
- MIT 6.006 — <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/>
- Abdul Bari — Algorithms — https://www.youtube.com/@abdul_bari
- Reference: CLRS, *Introduction to Algorithms*

Thread-T checkpoint: by the crescendo, implement from memory in C++ the core structures (dynamic arrays, linked lists, stacks/queues, hash maps, trees + BST, heaps, graph BFS/DFS) and the standard patterns (two-pointer, sliding window, binary search, backtracking, basic DP); solve a random LeetCode medium in ~30-35 min.

Thread M — Implement-by-hand ML (muscle memory) · ~2 hrs/week · highest-ROI thread

Directly validated: DeepMind's RE coding round asks you to implement model components/losses from a blank file (e.g. "implement Focal Loss from scratch in PyTorch" — one of your own domain losses). **This thread now absorbs cross-cutting habit #3 (Re-implementation Saturday)** — same activity, counted once. Each week, from a **blank file, no autocomplete, no AI**, reimplement one primitive in ~30-40 min, then revisit older ones spaced. The Phase-1 Karpathy build is the substrate.

Rotating list (cycle and revisit): backprop through an MLP · scaled dot-product + multi-head attention · a segmentation loss (Dice/Focal/Tversky) · cross-entropy from logits · LayerNorm/BatchNorm · softmax + stable log-softmax · a sampler (top-k/nucleus/temperature) · a DDPM step · logistic regression + gradient · k-means · PCA via SVD · a tiny SGD/Adam loop · positional encodings (sinusoidal/RoPE) · a KV-cache · **(v7) a message-passing GNN layer** · **(v7) translate one primitive to JAX.**

- Dive into Deep Learning — <https://d2l.ai/>
- labml.ai — annotated from-scratch PyTorch — <https://nn.labml.ai/>
- The Annotated Transformer — <http://nlp.seas.harvard.edu/annotated-transformer/>
- Karpathy micrograd / nanoGPT — <https://github.com/karpathy/micrograd> · <https://github.com/karpathy/nanoGPT>
- What the target rounds actually ask — <https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview>

Ramp: 1 primitive/week → implement key components of every capstone from scratch → in Phase 9, timed, unaided, interviewer-style drills. Because it's continuous, ramping toward an RS loop later is a gear-change, not new learning.

Thread R — Research output, networking & applications · ~1.5 hrs/week

The Tier-2/PhD differentiator and your pipeline engine. **This thread now absorbs cross-cutting habits #6 (reproduce-a-paper) and #7 (positioning doc)** — counted once. Rotating weekly:

- **Reproduce-a-target-paper (monthly):** publicly reproduce one paper from a target team (AlphaFold-lineage, DeepMind-science, medical imaging, geometric ML) and write it up. Triple duty: paper-discussion prep, referral hook, proof of level. Code: <https://paperswithcode.com/>
- **Write:** 1-2 short posts/month; maintain the "why me for digital biology" positioning narrative + two research job-talk decks.
- **Publish:** aim the spike capstone at a venue (see Publication strategy) — MICCAI <https://www.miccai.org/> · MIDL <https://midl.io/> · **MLSB** <https://www.mlsb.io/> · **LoG** <https://logconference.org/> · ISMB/RECOMB <https://www.iscb.org/> · preprints on arXiv <https://arxiv.org/> and bioRxiv <https://www.biorxiv.org/> · reviews via OpenReview <https://openreview.net/>
- **Applications:** run the company ladder (below); from Tier 0 (now) this thread carries your live loops.

BLOCK 1 — Depth, foundations & the spike

PHASE 0 — Rapid Brush-Up: Stats, ML & DL Fundamentals

~35 hrs (was 45) · reactivation for panel readiness · do this first · cures the stated pain point

Goal: rebuild crisp, *out-loud* command of the fundamentals so opening panel rounds never catch you flat. Reactivation is fast; the aim is fluent articulation, not re-derivation. **Weight the spoken drills** — that is the actual deliverable.

Primary resource — StatQuest: <https://www.youtube.com/@statquest/playlists> · <https://statquest.org/>

- **0.1 Statistics fundamentals (~9 hrs)** — distributions; mean/variance/SD; populations vs samples; CLT; sampling; hypothesis testing & p-values; CIs; R^2 ; covariance & correlation; MLE; bootstrapping; power. *Verbal checkpoint*: p-value, CI, CLT, MLE each in two sentences, no notes.
- **0.2 Classic ML (~14 hrs)** — bias-variance; train/val/test & CV; confusion matrix, sensitivity/specificity, precision/recall, ROC & AUC (and when PR is honest); linear & logistic regression; ridge vs lasso and why L1→sparsity; trees/RF/AdaBoost/gradient boosting/XGBoost; SVMs & the kernel trick; k-means & hierarchical clustering; PCA; naive Bayes; entropy & mutual info. *Verbal checkpoint, cold*: gradient boosting vs random forest; why L1 zeros weights and L2 shrinks them (at the gradient level); what AUC measures and when PR-AUC is more honest.
- **0.3 Deep-learning fundamentals (~8 hrs)** — StatQuest NN playlist + 3Blue1Brown NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi: perceptron/MLP; activations; backprop; gradient descent; losses; tensors; CNN & RNN/LSTM basics; overfitting, dropout, batch/layer norm, vanishing/exploding gradients; transfer learning.
- **0.4 Deliverable — panel articulation cheat-sheet (~4 hrs)** — one or two crisp spoken sentences per concept. Seeds the war-chest doc.

Phase 0 self-test (verbal, no notes): rapid-fire 15 of the above in 2–3 sentences each, out loud, as if to a panel.

PHASE 1 — Foundations: Mathematics + Neural Networks from First Principles

~120 hrs (was 190) · the bedrock — reactivation, not re-learning

Goal: mathematical maturity turned into working neural networks so backprop and "the why" are bone-deep. This is a *targeted refresh* for a daily user — watch the highest-leverage segments, not entire courses. By the end you can re-derive core results on paper and understand attention mechanically (Karpathy's GPT build), which unlocks Phase 2 and seeds Thread M.

- **1.1 Linear algebra (~35 hrs) (was 68 — the single biggest v6 over-estimate)**. Refresh to *teachable* fluency; you don't need all of 18.06 and 18.065.
 - 3B1B Essence of Linear Algebra — https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFtgF8hE_ab
 - MIT 18.06 (Strang) — *targeted lectures* — https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/
 - MIT 18.065 Matrix Methods (Strang) — *SVD/PCA/low-rank/GD sections only* — <https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/>
 - Reference: Matrix Cookbook — <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf>
 - *Whiteboard*: derive SVD from eigendecomposition; rank-r approximation geometrically; $A^T A$ is PSD; backprop through a linear layer with matrix calculus.
- **1.2 Calculus + matrix calculus (~15 hrs)**.
 - 3B1B Essence of Calculus — <https://www.youtube.com/playlist?list=PLZHQObOWTQDMSr9K-rj53DwVRMYO3t5Yr>
 - Matrix Calculus for Deep Learning (Parr & Howard) — <https://explained.ai/matrix-calculus/>
- **1.3 Probability & statistics (~25 hrs) (was 45)**. Targeted refresh — Stat 110 is large; hit the sections you're rusty on.
 - Harvard Stat 110 (Blitzstein) — <https://projects.iq.harvard.edu/stat110/youtube>
 - Alt: MIT RES.6-012 (Tsitsiklis) — <https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/>
 - *Whiteboard*: MLE for a Gaussian; conditional expectation as projection; Bayes on a non-trivial example.
- **1.4 Neural networks from first principles (~45 hrs) — the bridge · PROTECT this**. Do all 10 videos with exercises, re-implementing each without looking at his code. This is the substrate for Thread M and the highest-value part of Phase 1 — do not compress it.
 - Karpathy — Neural Networks: Zero to Hero — <https://www.youtube.com/playlist?list=PLAqhlrjkbxWl23v9cThsA9GvCAUhRvKZ> · hub <https://karpathy.ai/zero-to-hero.html>

Phase 1 self-test: hand-derive backprop for a 2-layer MLP; implement micrograd-style autodiff from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled dot-product attention and derive the $\sqrt{d_k}$ scaling. → **Threads T, M, R start now.**

PHASE 2 — Applied Computer Vision & Medical Image Segmentation

~110 hrs (was 185) · your production domain — this is **PACKAGING**, not re-learning · ends with Capstone 1 → Tier 0

Goal: convert what you already ship into *teachable, defensible, publicly-demonstrated* command — and produce the Tier-0 portfolio piece. You arrive understanding attention and backprop, so the loss math and architecture theses land immediately. **Apply to Tier-0 India medical-imaging roles in parallel with this phase — your eligibility exists now.**

- **2.1 Imaging physics for interviews (~12 hrs) (was 30)**. You work with these modalities; formalize the interview-ready story.
 - Allen Elster — Questions & Answers in MRI — <https://mriquestions.com/index.html>
 - OHBM Educational — MRI physics — <https://www.youtube.com/@ohbmonline/playlists>
 - Paul Callaghan — "Magnetic Resonance" (Otago), first 4 lectures — https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z
 - iBiology — fluorescence microscopy primer — <https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PlpdW9TMP>
 - *Deliverable*: 1-page note — T1/T2/FLAIR/T1CE + fluorescence artifacts (photobleaching, bleed-through, autofluorescence).
- **2.2 CNNs + convolution math + U-Net family + losses (~25 hrs) (was 50)**. Refresh the theory + the *implement-from-scratch* articulation (Thread M synergy).
 - CS231n 2017 (Johnson) — <https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLqRF3EO8sYv> · <https://cs231n.github.io/>
 - Convolution arithmetic (Dumoulin & Visin) — <https://arxiv.org/abs/1603.07285> · https://github.com/vdumoulin/conv_arithmetic
 - U-Net <https://arxiv.org/abs/1505.04597> · V-Net <https://arxiv.org/abs/1606.04797> · Attention U-Net <https://arxiv.org/abs/1804.03999> · nnU-Net <https://www.nature.com/articles/s41592-020-01008-z>
 - Losses (your 95/5 imbalance) — survey <https://arxiv.org/abs/2006.14822> · Generalized Dice <https://arxiv.org/abs/1707.03237> · Focal Tversky <https://arxiv.org/abs/1810.07842> · Boundary loss <https://arxiv.org/abs/1812.07032> (implement Dice/GDL/Focal/Tversky/Boundary/compound from scratch).
 - *Whiteboard*: U-Net with tensor shapes; why skip connections work; Dice gradient instability on empty masks; nnU-Net's "no new architecture" thesis.
- **2.3 nnU-Net + MONAI packaging (~15 hrs) (was 45)**. You already ship this — the work here is a *clean public reference*, not tutorials.

- MONAI tutorials — <https://github.com/Project-MONAI/tutorials> · videos <https://www.youtube.com/@projectmonai/videos>
- nnU-Net v2 source — <https://github.com/MIC-DKFZ/nnUNet> · MONAI integration <https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md> · Kitware comparison <https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/>
- DigitalSreeni — <https://www.youtube.com/@DigitalSreeni/playlists> · code https://github.com/bnsreenu/python_for_microscopists · U-Net playlist <https://www.youtube.com/playlist?list=PLZSOBAyNTZwbR08R959iCvYT3qzhxvGOE>
- AI for Medical Diagnosis (Rajpurkar) — audit free — <https://www.coursera.org/learn/ai-for-medical-diagnosis>
- **2.4 Transformer-based segmentation + foundation models (~15 hrs).** The genuinely newer pieces.
 - ViT <https://arxiv.org/abs/2010.11929> · TransUNet <https://arxiv.org/abs/2102.04306> · Swin-UNet <https://arxiv.org/abs/2105.05537> · UNETR <https://arxiv.org/abs/2103.10504> · Swin UNETR <https://arxiv.org/abs/2201.01266>
 - SAM <https://arxiv.org/abs/2304.02643> · MedSAM <https://www.nature.com/articles/s41467-024-44824-z> · MedNeXt <https://arxiv.org/abs/2303.09975>
- **2.5 3D, uncertainty, evaluation (~23 hrs) (was 40).**
 - TorchIO — <https://torchio.readthedocs.io/> · <https://www.youtube.com/playlist?list=PLfXIfDgnRPfIEHFx2N6mlbn-MGFOpIX2>
 - Uncertainty (Yarin Gal — MC dropout, ensembles, evidential) — https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning
 - Metrics Reloaded — <https://www.nature.com/articles/s41592-023-02151-z>
 - Domain shift (Stanford AIMI) — <https://aimi.stanford.edu/education/educational-resources>
 - 3D-conv exercise: input (D=64,H=128,W=128,C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 out-channels — output shape + param count by hand; repeat for transposed + dilated.

PHASE 2 — CAPSTONE 1 (~20 hrs) · public data, COI-safe · unlocks Tier 0 · the Tier-0 gate — fund it well. Notebook + 3-page write-up: (1) preprocessing with physics justification, (2) architecture ablation across 3 architectures, (3) loss-function ablation, (4) uncertainty + failure-mode analysis. Implement the losses and one architecture block from scratch (Thread M).

- **Datasets (pick one):** Medical Segmentation Decathlon (Task01 BrainTumour) <http://medicaldecathlon.com/> · BraTS <https://www.synapse.org/brats> · TCIA <https://www.cancerimagingarchive.net/> · IXI <https://brain-development.org/ixi-dataset/> — **or microscopy-native:** LIVECell <https://github.com/sartorius-research/LIVECell> · Sartorius (Kaggle) <https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation> · BBBC <https://bbbc.broadinstitute.org/>
- **Angle:** portfolio + a MIDL/MICCAI-workshop-grade short paper (Thread R). Directly Qure.ai / SigTuple / Google-India relevant.

Phase 2 self-test: whiteboard U-Net forward pass + skips with shapes; why nnU-Net usually beats fancier architectures (and when not); diagnose bias-field artifacts, class imbalance, label noise, domain shift on a new dataset.

PHASE 2B — Imaging & Video (life-sciences-native) (re-sequenced from v6 Phase 8)

~95 hrs · the fallback leg made modern · ends with Capstone 2 → strengthens Tier 1 (imaging)

Goal: extend your already-strong imaging leg into **video/temporal** — the closest genuine expansion of your existing strength, and the piece that unlocks phenomics/cell-imaging roles (Recursion) and strengthens the India case. This ran late (Phase 8) in v6; it belongs here, next to imaging, because it *compounds* your fallback strength rather than diverting into generic CV.

- **2B.1 Video & temporal analytics (~40 hrs).**
 - **Video transformers / action recognition:** VideoMAE <https://arxiv.org/abs/2203.12602> · TimeSformer <https://arxiv.org/abs/2102.05095> · ViViT <https://arxiv.org/abs/2103.15691> · SlowFast <https://arxiv.org/abs/1812.03982>
 - **Promptable video segmentation:** SAM 2 <https://arxiv.org/abs/2408.00714> · <https://github.com/facebookresearch/sam2> · <https://ai.meta.com/sam2> (extends your SAM/MedSAM work)
 - **Optical flow:** RAFT <https://arxiv.org/abs/2003.12039> · <https://github.com/princeton-vl/RAFT>
 - **Tracking (beyond DeepSORT):** ByteTrack <https://arxiv.org/abs/2110.06864> · <https://github.com/ifzhang/ByteTrack> · OC-SORT <https://arxiv.org/abs/2203.14360> · https://github.com/noahcao/OC_SORT · BoT-SORT <https://arxiv.org/abs/2206.14651> · <https://github.com/NirAharon/BoT-SORT>
 - **Metrics & benchmarks:** HOTA <https://arxiv.org/abs/2009.07736> · MOTChallenge <https://motchallenge.net/> · DAVIS <https://davischallenge.org/> · YouTube-VOS <https://youtube-vos.org/>
- **2B.2 Life-sciences-native video (~30 hrs) — the differentiator for the imaging leg.**
 - **Animal-behavior pose & tracking:** DeepLabCut <https://www.deeplabcut.org/> · <https://github.com/DeepLabCut/DeepLabCut> · Nature Neuroscience <https://www.nature.com/articles/s41593-018-0209-y> · SLEAP <https://sleap.ai/> · <https://github.com/talmolab/sleap> · Nature Methods <https://www.nature.com/articles/s41592-022-01426-1>
 - **Live-cell microscopy time-series:** Cellpose <https://www.cellpose.org/> · <https://github.com/MouseLand/cellpose> · TrackMate <https://imagej.net/plugins/trackmate/> · Ultrack <https://github.com/royerlab/ultrack> · Cell Tracking Challenge <http://celltrackingchallenge.net/>
- **2B.3 Detection/instance breadth (~25 hrs) — feeds cell-instance work.**
 - Michigan EECS 498 (Justin Johnson) — *detection/instance/panoptic sections* — <https://web.eecs.umich.edu/~justincj/teaching/eecs498/> · playlist <https://www.youtube.com/playlist?list=PL5-TkQAFzFbxjBHTzdVCWE0Zbhomg7r>

PHASE 2B — CAPSTONE 2 (video, ~included) · public data, COI-safe · strengthens Tier 1. DeepLabCut/SLEAP pose → temporal model → behavior classification; **or** Cellpose seg → TrackMate/Ultrack tracking + lineage; use SAM 2 for mask propagation.

- **Datasets (public):** CalMS21 <https://data.caltech.edu/records/s0vdx-0k302> · MABe <https://www.aicrowd.com/challenges/multi-agent-behavior-challenge-2022> · Cell Tracking Challenge <http://celltrackingchallenge.net/> · DAVIS/YouTube-VOS above.

Phase 2B self-test: SAM 2's memory mechanism for mask propagation; tracking-by-detection vs joint detection-and-tracking; when a video transformer beats frame-wise + tracking; the failure modes of cell-lineage tracking.

PHASE 3 — Deep Learning Theory & Modern Architectures

~105 hrs (was 150) · the broad, deep DL backbone · unlocks Tier 1 (broad)

Goal: generalize beyond Phase 1-2 — advanced transformer variants, self-supervised learning, generative/diffusion theory, training-dynamics math. Trim what you already use (transformers); *protect* the less-familiar generative/diffusion work — it feeds the relocation target's generative-for-molecules lane.

- **3.1 Optimization for deep learning (~18 hrs).** Boyd is large — targeted sections.
 - Sebastian Ruder — <https://www.ruder.io/optimizing-gradient-descent/>
 - Stanford EE364A Convex Optimization (Boyd) — *targeted* — <https://www.youtube.com/playlist?list=PL3940DD956CDF0622> · book <https://web.stanford.edu/~boyd/cvxbook/>
- **3.2 Information theory for ML (~12 hrs).**
 - David MacKay — ITILA, ch 1-6 — <https://www.inference.org.uk/itila/book.html>
 - *Whiteboard*: cross-entropy from KL; NLL = MLE; the ELBO; mutual information in InfoNCE.
- **3.3 Transformers in depth + modern variants (~25 hrs).** You'll have built a GPT already — this is variants.
 - Stanford CS25 — <https://web.stanford.edu/class/cs25/> · https://www.youtube.com/playlist?list=PLoROMvovd4rNijRchCzutFw5ltR_Z27CM
 - Lilian Weng — <https://lilianweng.github.io/> · Annotated Transformer — <http://nlp.seas.harvard.edu/annotated-transformer/>
 - RoPE <https://arxiv.org/abs/2104.09864> · RMSNorm <https://arxiv.org/abs/1910.07467> · FlashAttention <https://arxiv.org/abs/2205.14135> · GQA/MQA <https://arxiv.org/abs/2305.13245> · SwiGLU <https://arxiv.org/abs/2002.05202> · MoE/Switch <https://arxiv.org/abs/2101.03961>
- **3.4 Vision transformers + modern ConvNets (~18 hrs).**
 - ViT lectures (Lucas Beyer) — https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer
 - ConvNeXt <https://arxiv.org/abs/2201.03545> · DINOv2 <https://arxiv.org/abs/2304.07193> · MAE <https://arxiv.org/abs/2111.06377>
 - HF Computer Vision course — <https://huggingface.co/learn/computer-vision-course>
- **3.5 Generative modeling & diffusion (~32 hrs) · PROTECT — feeds the spike.**
 - Stanford CS236 (Ermon) — <https://www.youtube.com/playlist?list=PLoROMvovd4rPOWA-omMM6STXaWW4FvJT8>
 - DDPM <https://arxiv.org/abs/2006.11239> · Yang Song score-based <https://yang-song.net/blog/2021/score/> · Lilian Weng diffusion <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>
 - HF Diffusion course (units 1-2) — <https://huggingface.co/learn/diffusion-course>

Phase 3 self-test: derive the diffusion loss (variational + score-matching views); compare positional encodings; explain self-supervised pretraining in low-data regimes. → **Tier 1 (broader CV/DL roles).**

PHASE 4 — Quantum Computing & QML (*compressed, OFF the critical path*)

~100 hrs (was 275) · reactivation of an existing production strength · run this later/parallel, never blocking a target tier

Goal: match your production VQE/ADAPT-VQE/DMET-VQE experience with enough theory to *defend it to a skeptical chemist* and to *bridge into molecular ML* — not to become a quantum-computing specialist. The relocation targets hire on DL/GNN/generative; quantum's job here is (a) interview defense of your existing work and (b) the "quantum-informed" angle inside the Phase-6 spike capstone. **Full 275-hr version only if you target a quantum-specific role.**

Keep (the ~100 hrs): 4.4 (VQA math + barren plateaus), 4.5 (Schuld critique), the capstone-relevant slice of 4.6, and the minimum of 4.1-4.3 you need for those. *Defer unless targeting quantum roles:* the deep error-correction course and QDA breadth. All URLs kept regardless.

- **4.1 Group theory & symmetry (~10 hrs) — shared with the spike's equivariance.**
 - Group theory for physicists — https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures
 - Geometric Deep Learning lectures (Bronstein) — Lie-group/equivariance angle — <https://geometricdeeplearning.com/lectures/>
- **4.2 QM math: Hilbert spaces, operators, tensor products (~12 hrs).**
 - MIT 8.04 (Adams) — https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/
 - Quantum Country — <https://quantum.country/qcvc>
 - (Optional) Schuller — https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic
- **4.3 Quantum computing fundamentals (~20 hrs, targeted) (was 80).** Refresh, not a full re-run.
 - Basics of Quantum Information — <https://learning.quantum.ibm.com/course/basics-of-quantum-information> · YouTube <https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhVBHR>
 - Fundamentals of Quantum Algorithms — <https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms>
 - General Formulation — <https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information>
 - (*defer*) Foundations of Quantum Error Correction — <https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction>
 - Companion: Nielsen & Chuang, ch 1-4
- **4.4 Variational quantum algorithms — VQE, ADAPT-VQE, DMET, QAOA (~30 hrs) · KEEP.**
 - PennyLane Codebook — <https://pennylane.ai/codebook> · quantum-chemistry demos — https://pennylane.ai/qml/demos_quantum-chemistry
 - VQE <https://www.nature.com/articles/ncomms5213> · ADAPT-VQE <https://www.nature.com/articles/s41467-019-10988-2> · DMET-VQE <https://arxiv.org/abs/2108.08987> · QAOA <https://arxiv.org/abs/1411.4028> · Hadfield <https://arxiv.org/abs/1709.03489> · Pauli grouping <https://arxiv.org/abs/1907.13117>
- **4.5 Quantum machine learning theory (~25 hrs) · KEEP (the honest-take-on-hype material).**
 - PennyLane QML topic (Schuld-curated) — <https://pennylane.ai/topics/quantum-machine-learning>
 - Maria Schuld — "Taking stock of QML" — https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning
 - Schuld & Petruccione, ch 4-7
 - QML=kernels <https://arxiv.org/abs/2101.11020> · Barren plateaus <https://www.nature.com/articles/s41467-018-07090-4> · Expressibility <https://arxiv.org/abs/1905.10876> · Advantage-goal <https://arxiv.org/abs/2203.01340> · PQC encodings <https://arxiv.org/abs/2008.08605>
 - Courses to pull from: Péré <https://github.com/Christophe-pere/QML-Course> · Hjorth-Jensen <https://github.com/CompPhysics/QuantumComputi>

• 4.6 Quantum chemistry on quantum computers (~3 hrs refresh + folds into Capstone 3).

- Qiskit Global Summer School — <https://www.youtube.com/@qiskit/playlists>
- IBM Quantum Learning + Sample-based/Krylov Quantum Diagonalization — <https://learning.quantum.ibm.com/> · <https://www.ibm.com/quantum/blog/iql-migration>

The v6 standalone quantum capstone (old "Capstone 3b") is **absorbed into Capstone 3** as the "quantum-informed molecular ML" angle (delta-ML toward CCSD; DFT-vs-quantum data-generation), which advances the job target far more than a standalone VQE re-benchmark.

Phase 4 self-test: derive VQE from the variational principle; parameter-shift rule and why it works; the barren-plateau argument; VQE vs QAOA — when each and when *no* quantum algorithm; honest take on QML hype vs promise (cite Schuld); DMET fragmentation and the classical-quantum interface.

PHASE 5 — Full-Stack / Production AI Systems

~105 hrs (was 160) · production breadth + ML-systems-design readiness (in the RE loop)

Goal: upgrade from "I built it" to "I can defend every architectural choice and design the next-gen version." Trim CS336 (a semester-length build-from-scratch course that overlaps Phase 3 and Thread M) to targeted use; keep the ML-system-design core — it's a real RE round.

• 5.1 LLM internals (~20 hrs). CS336 *targeted*, not end-to-end.

- Stanford CS336 — <https://stanford-cs336.github.io/spring2024/> · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_
- HF NLP course — <https://huggingface.co/learn/nlp-course>

• 5.2 RAG at senior level (~18 hrs).

- Hamel Husain — <https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/>
- Eugene Yan — LLM patterns — <https://eugeneyan.com/writing/llm-patterns/>
- Anthropic — contextual retrieval — <https://www.anthropic.com/news/contextual-retrieval>
- LlamaIndex <https://docs.llamaindex.ai/> · RAGAS <https://docs.ragas.io/> · DSPy <https://dspace.ai/>

• 5.3 MLOps + ML systems design (~30 hrs) · KEEP (RE loop has ML system design).

- Chip Huyen — ML systems design — <https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html>
- Stanford CS329S — <https://stanford-cs329s.github.io/> · Made With ML — <https://madewithml.com/> · Full Stack DL — <https://fullstackdeeplearning.com/course/2022/>

• 5.4 Distributed training + inference optimization (~18 hrs).

- Sebastian Raschka — <https://magazine.sebastianraschka.com/>
- HF multi-GPU — https://huggingface.co/docs/transformers/en/perf_train_gpu_many
- vLLM <https://arxiv.org/abs/2309.06180> · quantization (GPTQ/AWQ/GGUF/FP8) · ZeRO <https://arxiv.org/abs/1910.02054>

• 5.5 Eval / observability / safety + ML-systems-design prep (~19 hrs).

- Anthropic — Constitutional AI — <https://www.anthropic.com/news/constitutional-ai> · Eugene Yan — LLM evals — <https://eugeneyan.com/writing/llm-evaluators/>
- Observability (LangSmith / Phoenix / Helicone) — run one on a real RAG system.
- ML-systems-design prep (not generic distributed systems): whiteboard, recorded, self-critiqued — "RAG assistant for HCPs, 10K queries/day on 50K docs"; "brain-seg pipeline ingesting new-institution data weekly with domain-shift handling"; "retraining + monitoring for a drug-discovery digital twin."

PHASE 6 — Geometric / Molecular ML — THE SPIKE

~175 hrs (was 150 — *the one phase to deepen*) · the differentiator for the relocation target + the PhD seed · ends with Capstone 3 (flagship) → Tier 2

Goal: close the geometric-deep-learning / molecular-AI gap that connects all your work — **the single highest-reward phase for the relocation target**. Isomorphic/DeepMind-science live in this lane, and your production quantum-chemistry background makes it *uniquely* credible: almost no one combines real VQE experience with equivariant-GNN capability. This is the "I can do what Isomorphic does, and I understand where the data comes from" story, on 100% public benchmarks.

• 6.1 Geometric deep learning + graph ML (~45 hrs). CS224W is large — take the GNN core (message passing, GraphSAGE, GAT, expressivity), skip the network-science/recommender weeks.

- Stanford CS224W (Leskovec) — <https://web.stanford.edu/class/cs224w/> · 2023 <https://snap.stanford.edu/class/cs224w-2023/> · playlist <https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxlpqhPgDQy7imNkDn>
- Bronstein et al. — GDL — <https://geometricdeeplearning.com/lectures/> · proto-book <https://arxiv.org/abs/2104.13478> · Veličković GNN lectures https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures
- Libraries: PyTorch Geometric <https://pytorch-geometric.readthedocs.io/> · DGL <https://www.dgl.ai/>

• 6.2 Molecular ML + equivariant networks (~55 hrs) · the core of the spike.

- DeepChem <https://deepchem.io/> · tutorials <https://deepchem.io/tutorials/> · Therapeutics Data Commons <https://tdcommons.ai/>
- SchNet <https://arxiv.org/abs/1706.08566> · DimeNet++ <https://arxiv.org/abs/2011.14115> · GemNet <https://arxiv.org/abs/2106.08903> · MolFormer <https://arxiv.org/abs/2106.09553> · equivariant nets (Cohen & Welling) <https://arxiv.org/abs/1602.07576>
- Equivariant NN stack:** EGNN <https://arxiv.org/abs/2102.09844> · e3nn <https://e3nn.org/> · <https://github.com/e3nn/e3nn> · tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP <https://github.com/mir-group/nequip> · paper <https://www.nature.com/articles/s41467-022-29939-5> · MACE <https://github.com/ACEsuit/mace> · paper <https://arxiv.org/abs/2206.07697> · Allegro <https://github.com/mir-group/allegro> · Equiformer <https://arxiv.org/abs/2206.11990> · curated list <https://github.com/Eipgen/Neural-Network-Models-for-Chemistry>

• 6.3 Protein structure + equivariance (~25 hrs).

- AlphaFold 2 — AlQuraishi explainer https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi · paper <https://www.nature.com/articles/s41586-021-03819-2>
- ESM/ESMFold <https://github.com/facebookresearch/esm> · ESM Atlas <https://esmatlas.com/> · Boltz <https://github.com/jwohlwend/boltz> · OpenFold <https://github.com/aqlaboratory/openfold>
- **6.4 JAX + mock interviews + portfolio (~20 hrs) (v7 adds light JAX).** Recorded, reviewed: 45-min segmentation deep-dive; 45-min quantum (defend VQE/ADAPT/DMET); 30-min behavioral (STAR, 4 stories). **Translate one capstone component to JAX** (relocation labs run JAX). Paper-of-the-week + write-ups (Thread R).
 - JAX docs — <https://jax.readthedocs.io/> · Equinox (JAX NN) — <https://docs.kidger.site/equinox/>

PHASE 6 — CAPSTONE 3 (~30 hrs, flagship) · public quantum-chemistry data, COI-safe · unlocks Tier 2 · the Isomorphic-differentiator piece. Train an **E(3)-equivariant GNN** (SchNet → DimeNet++ → EGNN/NequIP-style) to predict molecular properties; ablate equivariance; **fold in the quantum-informed angle** (delta-ML toward CCSD; discuss the DFT-vs-quantum-computing data-generation tie to your VQE work — this absorbs old Capstone 3b). Implement the message-passing core from scratch (Thread M); translate one component to JAX (6.4).

- **Datasets (public):** QM9 <https://quantum-machine.org/datasets/> · OGB-LSC PCQM4Mv2 <https://ogb.stanford.edu/kddcup2021/pcqm4m/> · MoleculeNet <https://moleculenet.org/> · OGB <https://ogb.stanford.edu/> · TDC <https://tdcommons.ai/>
- **Angle:** portfolio + an MLSB / AI4Science / LoG workshop paper (Thread R). The piece that says "I can do what Isomorphic does," on 100% public benchmarks. **This is the flagship — do it well.**

Phase 6 self-test: message passing vs convolution; what "equivariance" buys you and why it matters for molecules; when a GNN beats a fingerprint baseline; how AlphaFold's structure module uses equivariance; the quantum-chemistry data-generation story end-to-end.

BLOCK 2 — Breadth, GenAI & Research/Lab-Interview Readiness

~245 hrs + interviewing · Threads T + M + R continue underneath

PHASE 7 — Generative AI & Agentic AI (domain-focused, lighter)

~95 hrs (was 150) · competent, not specialist · off the Tier-0/Tier-2 critical path — run lighter/later · ends with optional Capstone 4

Goal: move from "I use LLMs in a RAG app" to *competent* design/adaptation/orchestration of modern GenAI/agentic systems at a domain-focused level — not a specialism. Keep this proportionate: it broadens you, it isn't the differentiator.

- **7.1 Advanced LLMs — adaptation & inference (~28 hrs).**
 - LoRA <https://arxiv.org/abs/2106.09685> · QLoRA <https://arxiv.org/abs/2305.14314> · Raschka's practical LoRA <https://magazine.sebastianraschka.com/>
 - InstructGPT/RLHF <https://arxiv.org/abs/2203.02155> · DPO <https://arxiv.org/abs/2305.18290> · HF TRL
 - HF LLM course — <https://huggingface.co/learn/llm-course>
 - Chip Huyen — *AI Engineering* (O'Reilly, 2025) — alongside.
- **7.2 Multimodal & vision-language models (~20 hrs) — also feeds imaging.**
 - CLIP <https://arxiv.org/abs/2103.00020> · LLaVA <https://arxiv.org/abs/2304.08485> · Flamingo <https://arxiv.org/abs/2204.14198>
- **7.3 Agents & agentic systems (~30 hrs) — competent level.**
 - HF Agents Course — <https://huggingface.co/learn/agents-course> · HF MCP Course — <https://huggingface.co/learn/mcp-course>
 - Anthropic — Building Effective Agents — <https://www.anthropic.com/research/building-effective-agents>
 - Andrew Ng — agentic design patterns — <https://www.deeplearning.ai/short-courses/> · LangGraph — <https://langchain-ai.github.io/langgraph/>
 - Patterns: ReAct, tool/function calling, reflection, planning/decomposition, memory, multi-agent; agent evaluation.

PHASE 7 — CAPSTONE 4 (~17 hrs, optional) · public literature, COI-safe. A multi-step research/analysis agent (RAG + tool use + memory + eval) over **open** scientific literature: "given a target/modality, retrieve → synthesize → self-critique." *Optional — do it only if you're targeting agent-heavy roles or have slack; it's not on the critical path.*

- **Corpora/tools (public):** PMC Open Access <https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/> · arXiv <https://arxiv.org/> · PubMed E-utilities <https://www.ncbi.nlm.nih.gov/books/NBK25501/> · Semantic Scholar API <https://www.semanticscholar.org/product/api> · structured tool-calling via TDC/ChEMBL. **Never ingest Lilly internal docs/data.**

Phase 7 self-test: explain LoRA and why it's parameter-efficient; RLHF vs DPO; when an agent beats a single well-prompted call; walk the ReAct loop; how you'd evaluate an agent's reliability.

PHASE 8 — CV-breadth literacy (classical / 3D / edge) — optional appendix

~30 hrs (was 230) · literacy only · NOT expertise, NOT positioned as seniority

Goal: enough *working literacy* to not freeze if a classical-vision question comes up — **not** a CV-expert build. The heavy classical/geometric/3D/SLAM/NeRF/edge-C++ content from v6 is de-prioritized: it's calibrated to the abandoned "generic senior CV engineer" goal and is ~irrelevant to life-sciences AI labs. The genuinely useful video/temporal material has moved to **Phase 2B**. Do this only with slack; never claim CV seniority in interviews.

- **8.1 Classical & geometric CV — literacy (~10 hrs).** Concepts only: camera models & calibration, homography, epipolar geometry, optical flow, SIFT/ORB.
 - First Principles of Computer Vision (Nayar) — *skim* — <https://fpcv.cs.columbia.edu/> · Coursera <https://www.coursera.org/specializations/firstprinciplesofcomputervision>
 - Cyrill Stachniss — <https://www.youtube.com/@CyrillStachniss>
 - Reference: Hartley & Zisserman (as needed)

- **8.2 Deployment/edge literacy (~15 hrs).** Enough to reason about serving, not a build.
 - ONNX <https://onnx.ai/> · NVIDIA TensorRT <https://developer.nvidia.com/tensorrt> · OpenCV <https://docs.opencv.org/> · PyImageSearch <https://pyimagesearch.com/> · CUDA basics (NVIDIA intro)
- **8.3 3D vision / neural rendering — awareness only (~5 hrs).**
 - NeRF <https://arxiv.org/abs/2003.08934> · 3D Gaussian Splatting <https://arxiv.org/abs/2308.04079>
 - Deploy datasets (if you go deeper on your own): KITTI <https://www.cvlibs.net/datasets/kitti/> · TUM RGB-D <https://cvg.cit.tum.de/data/datasets/rgbd-dataset> · ETH3D <https://www.eth3d.net/>

(v6's Capstone 5 "deployment" is dropped — it served the abandoned edge/integration framing. If you want a systems artifact, get Capstone 1/2/3's model running optimized behind an ONNX/TensorRT path as a short add-on, not a full capstone.)

Phase 8 self-test (literacy bar): explain the pinhole camera model and the epipolar constraint at a whiteboard; what TensorRT does to speed up inference; when 3D methods matter — enough to hold a conversation, not to claim expertise.

PHASE 9 — Research + Big-Tech Interview Crescendo

~115 hrs (was 150) · peak the interview-specific skills right as you interview · recalibrated to the RE loop, not a generic SWE loop · unlocks Tier 3

Goal: convert the year-long threads into performance. You're sharpening reflexes you already have. The RE loop at the target labs = **2 medium-hard coding rounds + ML fundamentals + ML system design + behavioral** (verified), plus a research discussion for research-adjacent roles — so this is weighted accordingly.

- **9.1 DSA — RE weight (~35 hrs) (was 55) — the crescendo of Thread T.** Timed NeetCode 150 → Blind 75 → company-tagged — <https://neetcode.io/> · <https://leetcode.com/> ; patterns under pressure — DesignGurus / Grokking <https://www.designgurus.io/> . Out loud, blank editor, ~35 min, **code must run** (CoderPad-style — a DeepMind quirk); ~3-4/day this phase. *Medium-hard fluency, not a hard-grind.*
- **9.2 Implement-by-hand ML (~35 hrs) — the crescendo of Thread M · highest value.** Timed, unaided, from a blank file: attention, a loss (Dice/Focal), a sampler, a diffusion step, a small training loop, backprop, **a message-passing layer**. *This is literally a DeepMind RE ML-round task.* Plus an ML math/theory sweep: gradient computation, KL properties, sampling, convergence intuition, the "L1 sparsity via gradients" mechanism (asked verbatim).
- **9.3 ML system design (~20 hrs) (was 30, de-genericized) — distinct from Phase 5.** Focus on ML system design (training/serving/data/eval/monitoring, domain shift), not generic distributed systems. ByteByteGo / Alex Xu for the general frame — <https://bytabytego.com/> · <https://www.youtube.com/@ByteByteGo> ; Gaurav Sen — <https://www.youtube.com/@gkcs> . (DDIA / MIT 6.824 optional and de-emphasized — they're SWE-system-design, low-yield for a science-lab RE.) Whiteboard 6-8 ML-flavored prompts, recorded, self-critiqued.
- **9.4 Research job-talk + behavioral (~25 hrs) · KEEP.** A 20-min talk + Q&A on each of your two strongest capstones (the segmentation/video piece and the molecular spike; expect constraints piled on iteratively). ~6-8 STAR stories mapped to each org's signals; low-level/OO design (parking lot, elevator, rate limiter) only for loops that include it; live peer mocks — Pramp / interviewing.io. **Research loops ban AI tools — rehearse fully unaided.**

Phase 9 self-test: a random LeetCode medium-hard in ~40 min with clean, running code; implement attention + a loss + a sampler + a GNN layer from a blank file, unaided, timed; a full ML system-design whiteboard in 45 min with capacity estimates and tradeoffs; a crisp 20-min job-talk on a capstone; four STAR stories from memory. → **Tier 3 (top research loops).**

(Active interviewing: keep DSA, implement-by-hand ML, and ML system design warm with ~4-5 hrs/week maintenance; use the buffer for on-sites, take-homes, and rest.)

Application overlay & company target ladder

You are already eligible at Tier 0 — enter the funnel **now**. Climb while the plan continues. Match the tier; treat early loops as calibration you get paid in feedback for. "Relocate" flags roles outside India. **Tiers re-pinned to the corrected critical path.**

Tier	Unlocked by	~When (sprint pace)	Apply to
Tier 0	Eligibility exists now; Capstone 1 strengthens	now → ~week 4-6	India medical-imaging AI + accessible; stretch Google-India
Tier 1	Capstone 1 + Capstone 2 (imaging + video)	~week 8-12	broader medical/biomedical imaging + video; stronger Google-India case
Tier 2	Capstone 3 (molecular GNN, flagship)	~week 12-18	AI drug-discovery labs; Isomorphic / DeepMind-science RE realistic
Tier 3	Interview crescendo (P9)	~week 20+	full top-lab loops

Tier 0 — India-domestic + accessible: Qure.ai (radiology imaging) <https://qure.ai/> · SigTuple (microscopy/cell — closest to your fluorescence work) <https://sigtuple.com/> · NVIDIA healthcare/MONAI + CUDA/TensorRT <https://www.nvidia.com/en-us/about-nvidia/careers/> · GE HealthCare (Bengaluru R&D) <https://www.gehealthcare.com/> · Microsoft Research India <https://www.microsoft.com/en-us/research/lab/microsoft-research-india/> · Google Research India (stretch now, real by Tier 1) <https://research.google/locations/india/>

Tier 1 — broaden: global medical-imaging AI if open to remote/relocation — Aidoc, Rad AI, Annalise.ai, PathAI, Paige.AI, Owkin (Paris/NYC, imaging + TechBio) <https://www.owkin.com/>

Tier 2 — AI drug discovery + research labs (relocate): Recursion (*phenomics = deep learning on cell images — a direct match to your medical-CV background applied to biology*) <https://www.recursion.com/careers> · Insitro <https://insitro.com/> · Genesis Therapeutics <https://www.genesis therapeutics.ai/> · Iambic <https://www.iambic.ai/> · Chai Discovery <https://www.chaidiscovery.com/> · EvolutionaryScale <https://www.evolutionaryscale.ai/> · Profluent <https://www.profluent.bio/> · InstaDeep (BioNTech; London/Paris) <https://www.instadeep.com/> · Valence Labs (Recursion's research arm, Montreal; great RS-track practice) <https://www.valencelabs.com/> · **Isomorphic Labs** <https://www.isomorphiclabs.com/careers> · **Google DeepMind** <https://deepmind.google/about/careers/>

- Target the **Research Engineer / ML Research Engineer** postings — verified to accept MSc + ~2 yrs ML model-development experience (no PhD). Save Research Scientist for after a PhD.

- *Neutral nuance*: Isomorphic has a large drug-discovery partnership with Eli Lilly — not a blocker (people move between partnered orgs), but apply with your normal discretion.

Tier 3 — top research loops: Isomorphic, Google DeepMind, Google Research (global + India), Genentech gRED / Prescient Design, Microsoft Research Health Futures, Meta FAIR.

Conflict-of-interest guardrails

Not legal advice — verify specifics with the right people at Lilly.

1. **Public data only**. Public benchmarks (Medical Decathlon/BraTS, QM9, PCQM4Mv2, CalMS21, PMC-OA, etc.). Never Lilly proprietary data, targets, molecules, assays, or internal docs. In Capstone 3 use textbook/public molecules (H₂/LiH/H₂O/QM9), **not** your Lilly molecules (avoid the keto-enol / HATU work).
2. **Generic, published methods** in any portfolio artifact — no Lilly trade secrets or unpublished internal techniques.
3. **Personal time and resources** — your own hardware/accounts; not Lilly compute, licenses, or work hours. (*Note: any work on Lilly hardware/networks must use Lilly's JFrog Artifactory for packages per enterprise policy — but these capstones are explicitly personal-hardware/personal-account, outside that environment, where standard package sources apply.*)
4. **Clear before you publish** — run any paper/blog/repo/talk through Lilly's standard external-publication / IP-review process; when unsure, ask first.
5. **Check your agreement** — review employment / invention-assignment / moonlighting terms; confirm portfolio work is clear.
6. **Positioning stays generic** — your "why me" narrative is about transferable skills, never Lilly specifics.

Cross-cutting habits (every week, no exceptions)

*De-duplicated in v7: the old habits #3, #6, #7 are now **inside** Threads M and R so nothing is double-counted.*

1. **Paper-of-the-week** — one paper, deeply read, derivations redone on paper. (Supersedes the unrealistic "1 paper/day.") Don't break the streak.
2. **Whiteboard Friday** — one concept whiteboarded without notes; photograph it. These become your interview-prep deck.
3. **Sunday writeup** — a 1-page note explaining the week's deepest concept to a hypothetical junior. Teaching is the senior-level forcing function.
4. **Senior-engineer "war chest"** — one Obsidian/Notion doc collecting derivations, paper one-liners, gotchas, anti-patterns. Your interview safety net.

*(Re-implementation Saturday → now **Thread M**. Reproduce-a-target-paper + the living positioning doc → now **Thread R**.)*

PhD-later decision guide (new in v7)

Whether. For the long-term **Senior-RS** aspiration, a PhD is effectively required at the target labs: Isomorphic's *Research Scientist* posting asks for a PhD + a publication track record, while its *Research Engineer* posting explicitly accepts MSc + ~2 yrs ML experience (no PhD). So the PhD question is really: *do you want the RS title, or is a senior/staff RE track enough?* If RE/staff suffices, the PhD is genuinely optional.

When. After landing a strong RE role — job-first materially expands your options.

Where (and how job-first improves each):

- **India — part-time / external (moonlight-compatible)**. IISc's **External Registration Programme** and the IITs' external/part-time PhD tracks admit working professionals (~2 yrs' experience; IISc needs ~1 yr on-campus + a company co-advisor; part-time runs ~6 yrs). Best if you land at an India R&D employer (Google Research India, MSR India) that can co-sponsor. — <https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/>
- **EU — salaried / industrial (a career move, not a moonlight)**. NL/Germany/Sweden/Switzerland/Austria PhDs are *salaried employee* roles (~€3–6k/mo, ~4 yrs); **MSCA Industrial Doctoral Networks** embed the PhD across a company + university. The strongest *prestige-and-funding* leap over a self-funded tier-2/3 Indian PhD — and a strong industry portfolio + a first-author workshop paper + an advisor relationship is exactly what makes it competitive. — <https://euraxess.ec.europa.eu/>
- **USA — mostly full-time funded; part-time rare at top CS programs**. More realistic as a *research-embedded* path (reach a lab, then collaborate/sponsor into a PhD) — or find that senior/staff RE makes the PhD unnecessary.

What to do now (zero marginal cost, no slowdown to the job hunt): the spike capstone → a **non-archival workshop paper** + a **bioRxiv/arXiv preprint**; build **1-2 advisor relationships** by reproducing a target lab's work, engaging at the workshop, and emailing thoughtfully; keep the geometric/structural-bio identity a PhD would extend. Every one of these is *already* produced by Thread R + the spike.

Decision criteria when the time comes (priority order): advisor fit → topic continuity with your spike → funding model (salaried/industrial > self-funded) → part-time feasibility vs your job → location/visa → program prestige. *Prestige is last*: a funded EU/industrial PhD under a strong advisor beats a higher-ranked program that is a poor topical or funding fit.

Publication & portfolio strategy (new in v7)

Realistic output for a working professional over ~12-18 months (be honest): **1-2 non-archival workshop papers + 1 preprint + clean open-source repos** — not a solo top-conference main-track paper.

- **Job lever (near-term)**. Reproduce-a-paper + a workshop paper = paper-discussion prep, a warm referral hook, and third-party proof of level for RE loops. MLSB explicitly prioritizes submissions with **open-sourced code** for spotlights — ship the repo clean; the code *is* the credential.
- **PhD-optionality lever**. First-author workshop papers + preprints + an advisor relationship are the exact portfolio that lifts you from "self-

funded tier-2/3 PhD" to "funded EU/industrial or research-embedded PhD."

- **Venues & realistic timing.** Imaging/video capstone → **MIDL / MICCAI workshop** (spring/summer) — <https://midl.io/> · <https://www.miccai.org/> · Molecular capstone → **MLSB** (NeurIPS workshop, ~Sep/Oct submission, Dec) — <https://www.mlsb.io/> · **LoG** (late autumn) — <https://logconferece.org/> · **AI4Science / Simbiochem** (NeurIPS). *MLSB-2026's window is ~now — likely too soon for a fresh spike — so target MIDL-2027 / MLSB-2027 + a rolling bioRxiv/arXiv preprint you can cite immediately.*
- **Visibility levers:** open-source every capstone with a reproducible README; reproduce a target lab's result and publish the write-up; contribute a small, real fix to an ecosystem repo (**e3nn / MACE / PyG / MONAI**) — a merged PR into a lab's own stack beats another blog post.

Risk register (new in v7)

Risk	Like liho od	Im pa ct	Leading indicator	Mitigation
Burnout (the #1 risk at 45 hrs/wk on top of full-time)	High	Hi gh	Skipped rest blocks; capstone quality dropping; dread of sessions	Sprint the critical path, then throttle to ~25-30 hrs/wk; threads are shock-absorbers; protect one true rest day
Breadth-without-depth	High	Hi gh	Can use a topic but can't derive/teach it at a whiteboard	Self-tests are the gate, not the calendar; Thread M + Whiteboard Friday enforce depth; cut P7/legacy-P8 to protect the spike
Over-claiming CV seniority	Med	Hi gh	CV/positioning says "senior CV engineer"; you freeze on classical-vision/SLAM questions	Reposition as "medical-imaging + geometric/molecular ML"; claim CV breadth as <i>working knowledge</i> ; the retitle
Unrealistic hours → schedule slip	Med	Me d	Consistently behind the phase self-tests	Corrected per-phase hours + flexible pace + self-tests-as-gates; dates are aids
Goal drift (the plan's own failure mode)	High	Me d	Adding "just one more" course; re-bloating	Fix the two-legged identity; every addition must serve imaging-video or molecular-geometric or the RE loop, else cut
Relocation-vs-domestic split	Med	Me d	Effort split 50/50, mediocre at both	The two legs share an umbrella so effort compounds; apply Tier-0 India <i>now</i> while building the Tier-2 spike; let the market decide
Quantum sunk-cost	Med	Me d	Pouring 275 hrs into quantum at the spike's expense	Compressed to ~100 hrs, off the critical path; quantum serves only as the molecular-ML bridge
Premature / ill-fitting PhD	Low-Med	Hi gh	Applying to a self-funded tier-2/3 PhD before landing the job	PhD is post-job; job-first expands options; decision criteria applied

Resource index (quick lookup) — all v6 URLs preserved

Brush-up (P0) — StatQuest <https://www.youtube.com/@statquest/playlists> · <https://statquest.org/> · 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Foundations math (P1) — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · 3B1B Calc <https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr> · MIT 18.06 <https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/> · MIT 18.065 <https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/> · Stat 110 <https://projects.iq.harvard.edu/stat110/youtube> · MIT RES.6-012 <https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/> · Matrix Cookbook <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf> · Matrix Calculus <https://explained.ai/matrix-calculus/> · Karpathy Z2H <https://www.youtube.com/playlist?list=PLAqhlrjkbxWl23v9cThsA9GvCAUhRvKZ> · hub <https://karpathy.ai/zero-to-hero.html>

Later-threaded math — Boyd <https://www.youtube.com/playlist?list=PL3940DD956CDF0622> · <https://web.stanford.edu/~boyd/cvxbook/> · Rudier <https://www.rudier.io/optimizing-gradient-descent/> · MacKay <https://www.inference.org.uk/itila/book.html> · Group theory https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures <https://geometricdeeplearning.com/lectures/>

Medical imaging (P2) — MRI Q&A <https://mriquestions.com/index.html> · OHBM <https://www.youtube.com/@ohbmonline/playlists> · Callaghan http://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology <https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhWfX6jMI3PlpdW9TMP> · AI for Medical Diagnosis <https://www.coursera.org/learn/ai-for-medical-diagnosis> · DigitalSreeni <https://www.youtube.com/@DigitalSreeni/playlists> · code https://github.com/bnsreenu/python_for_microscopists · Conv arithmetic <https://arxiv.org/abs/1603.07285> · https://github.com/vdumoulin/conv_arithmetic · MONAI <https://github.com/Project-MONAI/tutorials> · <https://www.youtube.com/@projectmonai/videos> · nnU-Net <https://github.com/MIC-DKFZ/nnUNet> · TorchIO <https://torchio.readthedocs.io/> · Stanford AIMI <https://aimi.stanford.edu/education/educational-resources> · CS231n <https://www.youtube.com/playlist?list=PL3FW7Lu3i5jvHM8ljY-zLfQRF3EO8sYv> · <https://cs231n.github.io/> · Metrics Reloaded <https://www.nature.com/articles/s41592-023-02151-z> · Seg papers: U-Net <https://arxiv.org/abs/1505.04597> · V-Net <https://arxiv.org/abs/1606.04797> · Attention U-Net <https://arxiv.org/abs/1804.03999> · nnU-Net <https://www.nature.com/articles/s41592-020-01008-z> · Loss survey <https://arxiv.org/abs/2006.14822> · Generalized Dice <https://arxiv.org/abs/1707.03237> · Focal Tversky <https://arxiv.org/abs/1810.07842> · Boundary loss <https://arxiv.org/abs/1812.07032> · ViT <https://arxiv.org/abs/2010.11929> · TransUNet <https://arxiv.org/abs/2102.04306> · Swin-UNet <https://arxiv.org/abs/2105.05537> · UNETR <https://arxiv.org/abs/2103.10504> · Swin UNETR <https://arxiv.org/abs/2201.01266> · SAM <https://arxiv.org/abs/2304.02643> · MedSAM <https://www.nature.com/articles/s41467-024-44824-z> · MedNeXt <https://arxiv.org/abs/2303.09975> · **Datasets:** Medical Decathlon <http://medicaldecathlon.com/> · BraTS <https://www.synapse.org/brats> · TCIA <https://www.cancerimagingarchive.net/> · IXI <https://brain-development.org/ixi-dataset/> · LIVECell <https://github.com/sartorius-research/LIVECell> · Sartorius Kaggle <https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation> · BBC <https://bbbc.broadinstitute.org/>

Deep learning core (P3) — CS25 <https://web.stanford.edu/class/cs25/> · https://www.youtube.com/playlist?list=PLoROMvovd4rNijRchCzutFw5ItR_Z27CM · CS336 <https://stanford-cs336.github.io/spring2024/> · https://www.youtube.com/playlist?list=PLoROMvovd4rOY23Y0BoGoBGgQ1zmU_MT_ · CS236 <https://www.youtube.com/playlist?list=PLoROMvovd4rPOWA-omMM6STXaWW4FvJT8> · Lilian Weng <https://lilianweng.github.io/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · HF NLP <https://huggingface.co/learn/nlp-course> · HF CV <https://huggingface.co/learn/computer-vision-course> · HF Diffusion <https://huggingface.co/learn/diffusion-course> · RoPE <https://arxiv.org/abs/2104.09864> · RMSNorm <https://arxiv.org/abs/1910.07467> · FlashAttention <https://arxiv.org/abs/2205.14135> · GQA <https://arxiv.org/abs/2305.13245> · SwiGLU <https://arxiv.org/abs/2002.05202> · Switch/MoE <https://arxiv.org/abs/2101.03961> · ConvNeXt <https://arxiv.org/abs/2201.03545> · DINOv2 <https://arxiv.org/abs/2304.07193> · MAE <https://arxiv.org/abs/2111.06377> · DDPM <https://arxiv.org/abs/2006.11239> · Score-based <https://yang-song.net/blog/2021/score/> · Diffusion

(blog) <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

Quantum (P4) — IBM Quantum Learning <https://learning.quantum.ibm.com/> · Watrous <https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTM y5Y2lqmSaUjfnhvBHR> · Basics <https://learning.quantum.ibm.com/course/basics-of-quantum-information> · Algorithms <https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms> · General Formulation <https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information> · Error Correction <https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction> · Qiskit <https://www.youtube.com/@qiskit/playlists> · QDA blog <https://www.ibm.com/quantum/blog/iql-migration> · PennyLane codebook <https://pennylane.ai/codebook> · PennyLane QML <https://pennylane.ai/topics/quantum-machine-learning> · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · MIT 8.04 <https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/> · Quantum Country <https://quantum.country/qcvc> · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic · Péré <https://github.com/Christophe-pere/QML-Course> · Hjorth-Jensen <https://github.com/CompPhysics/QuantumComputingMachineLearning> · VQE <https://www.nature.com/articles/ncomms5213> · ADAPT-VQE <https://www.nature.com/articles/s41467-019-10988-2> · DMET-VQE <https://arxiv.org/abs/2108.08987> · QAOA <https://arxiv.org/abs/1411.4028> · Hadfield <https://arxiv.org/abs/1709.03489> · QML=kernels <https://arxiv.org/abs/2101.11020> · Barren plateaus <https://www.nature.com/articles/s41467-018-07090-4> · Expressibility <https://arxiv.org/abs/1905.10876> · Advantage-goal <https://arxiv.org/abs/2203.01340> · PQC encodings <https://arxiv.org/abs/2008.08605> · Pauli grouping <https://arxiv.org/abs/1907.13117>

Production / MLOps (P5) — CS329S <https://stanford-cs329s.github.io/> · Made With ML <https://madewithml.com/> · Full Stack DL <https://fullstackdeeplearning.com/course/2022/> · Chip Huyen <https://huyenchip.com/ml-interviews-book/> · Hamel Husain <https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/> · Eugene Yan patterns <https://eugeneyan.com/writing/llm-patterns/> · Eugene Yan evals <https://eugeneyan.com/writing/llm-evaluators/> · Anthropic contextual retrieval <https://www.anthropic.com/news/contextual-retrieval> · Anthropic Constitutional AI <https://www.anthropic.com/news/constitutional-ai> · LlamaIndex <https://docs.llamaindex.ai/> · RAGAS <https://docs.ragas.io/> · DSPy <https://dspy.ai/> · Raschka <https://magazine.sebastianraschka.com/> · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM <https://arxiv.org/abs/2309.06180> · ZeRO <https://arxiv.org/abs/1910.02054>

DSA & C++ (Thread T) — learncpp <https://www.learncpp.com/> · The Cherno <https://www.youtube.com/@TheCherno> · cppreference <https://en.cppreference.com/> · NeetCode <https://neetcode.io/> · LeetCode <https://leetcode.com/> · MIT 6.006 <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/> · Abdul Bari https://www.youtube.com/@abdul_bari

Implement-by-hand ML (Thread M) — d2l.ai <https://d2l.ai/> · labml.ai <https://nn.labml.ai/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · micrograd <https://github.com/karpathy/micrograd> · nanoGPT <https://github.com/karpathy/nanoGPT> · DeepMind RE round guide <https://igotanooffer.com/en/advice/google-deepmind-research-engineer-interview> · **JAX (new):** <https://jax.readthedocs.io/> · Equinox <https://docs.kidger.site/equinox/>

Research output (Thread R) — Papers with Code <https://paperswithcode.com/> · MICCAI <https://www.miccai.org/> · MIDL <https://midl.io/> · MLSB <https://www.mlsb.io/> · LoG <https://logconference.org/> · ISCB/ISMB <https://www.iscb.org/> · arXiv <https://arxiv.org/> · bioRxiv <https://www.biorxiv.org/> · OpenReview <https://openreview.net/>

GenAI & Agents (P7) — HF LLM course <https://huggingface.co/learn/llm-course> · HF Agents <https://huggingface.co/learn/agents-course> · HF MCP <https://huggingface.co/learn/mcp-course> · Anthropic Building Effective Agents <https://www.anthropic.com/research/building-effective-agents> · DeepLearning.AI short courses <https://www.deeplearning.ai/short-courses/> · LangGraph <https://langchain-ai.github.io/langgraph/> · LoRA <https://arxiv.org/abs/2106.09685> · QLoRA <https://arxiv.org/abs/2305.14314> · InstructGPT/RLHF <https://arxiv.org/abs/2203.02155> · DPO <https://arxiv.org/abs/2305.18290> · CLIP <https://arxiv.org/abs/2103.00020> · LLaVA <https://arxiv.org/abs/2304.08485> · Flamingo <https://arxiv.org/abs/2204.14198> · **Corpora:** PMC-OA <https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/> · PubMed E-utilities <https://www.ncbi.nlm.nih.gov/books/NBK25501/> · Semantic Scholar API <https://www.semanticscholar.org/product/api>

Video & imaging (P2B) + optional CV literacy (P8) — First Principles of CV <https://fpcv.cs.columbia.edu/> · FPCV Coursera <https://www.coursera.org/specializations/firstprinciplesofcomputervision> · Cyrill Stachniss <https://www.youtube.com/@CyrillStachniss> · Michigan EECS 498 <https://web.eecs.umich.edu/~justincj/teaching/eecs498/> · playlist <https://www.youtube.com/playlist?list=PL5-TkQAFzFbzxjBHTzdVCWE0Zbhong7r> · NeRF <https://arxiv.org/abs/2003.08934> · 3D Gaussian Splatting <https://arxiv.org/abs/2308.04079> · OpenCV <https://docs.opencv.org/> · PyImageSearch <https://pyimagesearch.com/> · ONNX <https://onnx.ai/> · TensorRT <https://developer.nvidia.com/tensorrt> · **Video:** VideoMAE <https://arxiv.org/abs/2203.12602> · TimeSformer <https://arxiv.org/abs/2102.05095> · ViViT <https://arxiv.org/abs/2103.15691> · SlowFast <https://arxiv.org/abs/1812.03982> · SAM 2 <https://arxiv.org/abs/2408.00714> · <https://github.com/facebookresearch/sam2> · RAFT <https://arxiv.org/abs/2003.12039> · <https://github.com/princeton-vl/RAFT> · ByteTrack <https://arxiv.org/abs/2110.06864> · <https://github.com/ifzhang/ByteTrack> · OC-SORT <https://arxiv.org/abs/2203.14360> · https://github.com/noahcao/OC_SORT · BoT-SORT <https://arxiv.org/abs/2206.14651> · <https://github.com/NirAharon/BoT-SORT> · HOTA <https://arxiv.org/abs/2009.07736> · MOTChallenge <https://motchallenge.net/> · DAVIS <https://davischallenge.org/> · YouTube-VOS <https://youtube-vos.org/> · **Life-sciences video:** DeepLabCut <https://www.deeplabcut.org/> · SLEAP <https://sleap.ai/> · Cellpose <https://www.cellpose.org/> · TrackMate <https://imagej.net/plugin/trackmate/> · UltracTrack <https://github.com/royerlab/ultrack> · Cell Tracking Challenge <http://celltrackingchallenge.net/> · CalMS21 <https://data.caltech.edu/records/s0vdx-0k302> · **Deploy datasets:** KITTI <https://www.cvlibs.net/datasets/kitti/> · TUM RGB-D <https://cvg.cit.tum.de/data/datasets/rgbd-dataset> · ETH3D <https://www.eth3d.net/>

Geometric / Molecular ML (P6 — the spike) — CS224W <https://web.stanford.edu/class/cs224w/> · <https://snap.stanford.edu/class/cs224w-2023/> · playlist <https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxlpqhjhPgDQy7imNkDn> · Bronstein GDL <https://geometricdeeplearning.com/lectures/> · proto-book <https://arxiv.org/abs/2104.13478> · PyG <https://pytorch-geometric.readthedocs.io/> · DGL <https://www.dgl.ai/> · DeepChem <https://deepchem.io/tutorials/> · TDC <https://tdcommons.ai/> · SchNet <https://arxiv.org/abs/1706.08566> · DimeNet++ <https://arxiv.org/abs/2011.14115> · GemNet <https://arxiv.org/abs/2106.08903> · MolFormer <https://arxiv.org/abs/2106.09553> · Equivariant (Cohen&Welling) <https://arxiv.org/abs/1602.07576> · EGNN <https://arxiv.org/abs/2102.09844> · e3nn <https://e3nn.org/> · <https://github.com/e3nn/e3nn> · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP <https://github.com/mir-group/nequip> · <https://www.nature.com/articles/s41467-022-29939-5> · MACE <https://github.com/ACEsuit/mace> · <https://arxiv.org/abs/2206.07697> · Allegro <https://github.com/mir-group/allegro> · Equiformer <https://arxiv.org/abs/2206.11990> · curated list <https://github.com/Eipgen/Neural-Network-Models-for-Chemistry> · AlphaFold 2 <https://www.nature.com/articles/s41586-021-03819-2> · ESM <https://github.com/facebookresearch/esm> · ESM Atlas <https://esmatlas.com/> · Boltz <https://github.com/jwohlwend/boltz> · OpenFold <https://github.com/aqlaboratory/openfold> · **Datasets:** QM9 <https://quantum-machine.org/datasets/> · PCQM4Mv2 <https://ogb.stanford.edu/kddcup2021/pcqm4m/> · MoleculeNet <https://moleculenet.org/> · OGB <https://ogb.stanford.edu/>

Interview (P9) — NeetCode <https://neetcode.io/> · LeetCode <https://leetcode.com/> · DesignGurus <https://www.designgurus.io/> · ByteByteGo <https://bytebytego.com/> · <https://www.youtube.com/@ByteByteGo> · Gaurav Sen <https://www.youtube.com/@gkcs> · MIT 6.824 (optional/de-emphasized) <https://pdos.csail.mit.edu/6.824/>

Company ladder — Qure.ai <https://quire.ai/> · SigTuple <https://sigtuple.com/> · NVIDIA <https://www.nvidia.com/en-us/about-nvidia/careers/> · GE HealthCare <https://www.gehealthcare.com/> · MSR India <https://www.microsoft.com/en-us/research/lab/microsoft-research-india/> · Google Research

India <https://research.google/locations/india/> · Owkin <https://www.owkin.com/> · Recursion <https://www.recursion.com/careers> · Insitro <https://insitro.com/> · Genesis <https://www.genesistherapeutics.ai/> · Iambic <https://www.iambic.ai/> · Chai <https://www.chaidiscovery.com/> · EvolutionaryScale <https://www.evolutionaryscale.ai/> · Profluent <https://www.profluent.bio/> · InstaDeep <https://www.instadeep.com/> · Valence Labs <https://www.valencelabs.com/> · Isomorphic Labs <https://www.isomorphiclabs.com/careers> · Google DeepMind <https://deepmind.google/about/careers/> · **PhD routes (new)**: IISc ERP <https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/> · EURAXESS <https://euraxess.ec.europa.eu/>

A note on what you're taking on

This is now a **~1,360-hour** program on top of a full-time role — still ambitious, but honestly scoped and front-loaded for early wins. Anchors: the self-tests and capstones are the real gates, not the calendar; the three threads are your shock absorbers — dial them to ~2-3 hrs/week combined whenever a week gets heavy. **Apply from Tier 0 now** — your eligibility already exists in your production domain; the plan makes you better mid-interview, so don't pause it. Sprint the critical path at your 45-hr capacity, then **throttle to ~25-30 hrs/week** — burnout in month five costs more than any schedule it appears to save. Re-activating rusty fundamentals will feel like going backwards at first — it isn't; it's what turns "I use this" into "I can teach this to a panel." And spend your freed hours on the **spike**, not the legacy: geometric/molecular ML bridged from your quantum chemistry is the one thing that makes you *uniquely* hireable for the relocation target and seeds a later PhD. Adjust as you go — it's a scaffold for durable mastery and a real, early shot at a life-sciences research-engineering seat, sequenced so you don't wait a year to start competing.